Learning Curve Dynamics of Gradient Boosting Decision Trees on Small Datasets: Non-monotonic Overfitting Gaps and Metric Convergence

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Abstract

We investigate how classifier performance and the overfitting gap (training score minus validation score) scale with training set size on the UCI Wine dataset, a small (178-sample) structured benchmark. We train a Gradient Boosting Decision Tree (GBDT) classifier and an MLP baseline across ten logarithmically spaced training fractions (5%–95% of a 142-sample training pool), repeated ten times per fraction with different sub-sampling seeds. Our main findings are as follows. (1) The GBDT overfitting gap decreases overall from 0.114 to 0.028, though the validation accuracy curve is non-monotonic, dipping to 0.869 at 13% training fraction before recovering to 0.972 at 95%. (2) The macro-F1 gap closes at a faster rate (0.155) than the accuracy gap (0.124) per unit of log-training-size, suggesting F1 is more sensitive to improving class-level generalisation. (3) Hypothesis H3—that the gap inflection point falls between 40–60% of the training pool—is clearly falsified: the empirical inflection occurs at approximately 25.7%. (4) Both GBDT and MLP exhibit the same number of sign changes in the discrete second derivative of the gap curve (three each), suggesting that small-sample stochasticity inherent to this 178-sample dataset, rather than GBDT-specific ensemble dynamics, is the primary driver of non-monotonic behaviour. (5) GBDT wall-clock training time increases by a factor of 2.23 over a 19× increase in training fraction, contradicting the super-linear scaling prediction. All quantitative claims are empirical observations; formal statistical testing is limited by the small sample size, and we discuss implications and limitations accordingly.

1 Introduction

Understanding how classifier performance scales with training set size is fundamental to practical machine learning, particularly for small structured datasets where data acquisition is expensive [**?**]. Learning curves—plots of training and validation performance as a function of training set size—reveal whether a model is in a high-bias or high-variance regime and guide decisions about when additional data collection will yield diminishing returns [**?**].

Gradient Boosting Decision Trees (GBDT) [**?**] are an ensemble method that builds additive models of shallow decision trees, sequentially correcting residual errors from preceding trees. On structured, numerical datasets such as the UCI Wine dataset [**?**], GBDT frequently achieves state-of-the-art accuracy [**?**]. However, the interaction between GBDT's ensemble hyperparameters (tree depth, number of estimators, learning rate) and the amount of available training data introduces a complex bias–variance trade-off that is not fully characterised by scalar accuracy metrics alone [**?**].

Prior learning curve studies for GBDT predominantly report a single scalar metric (most commonly accuracy) and treat the overfitting gap as a monotonically decreasing function of training size [**?**]. This framing obscures two potentially important phenomena. First, the gap trajectory may be non-monotonic: as the training set grows from very small sizes, the ensemble may pass through a regime in which additional data initially increases misspecification before the variance finally collapses [**?**]. Second, macro-F1 and raw accuracy may converge

to their respective asymptotes at different rates because F1 penalises class-imbalance-induced overconfidence that persists longer in accuracy [**?**].

We also note a well-documented limitation of MLP-style neural networks on structured datasets: prior work has shown that neural networks can underperform tree-based methods on structured or tabular data due to their inductive biases being better suited to spatial or sequential structure [**?**]. We acknowledge this limitation explicitly when interpreting the GBDT–MLP comparison; the MLP is used here not as an optimal competitor but as a falsification control to assess whether observed gap behaviour is GBDT-specific.

Building on these observations, we test five pre-registered hypotheses:

(H1) The GBDT overfitting gap decreases overall as training set size increases.

(H2) The macro-F1 overfitting gap converges to its asymptote faster than the accuracy overfitting gap.

(H3) The critical inflection point of the GBDT gap curve falls between 40–60% of the training pool.

(H4) The GBDT gap exhibits greater non-monotonicity than the MLP baseline gap, attributing the effect to GBDT ensemble dynamics rather than dataset characteristics.

(H5) GBDT wall-clock training time scales super-linearly with training set size.

The remainder of the paper is structured as follows. Section 2 describes the dataset, models, experimental protocol, and evaluation metrics. Section 3 reports the learning curve measurements and hypothesis test outcomes, along with discussion of statistical limitations. Section 4 summarises implications for practitioners.

2 Methods

2.1 Dataset

We use the UCI Wine dataset [**?**] as distributed through `sklearn.datasets.load_wine` [**?**]. The dataset comprises 178 samples described by 13 continuous chemical-measurement features and labelled with one of three wine cultivar classes. The class distribution is mildly imbalanced: 59, 71, and 48 samples for classes 0, 1, and 2 respectively. All features are real-valued measurements (e.g., alcohol content, malic acid, flavanoids), making the dataset amenable to tree-based methods without categorical encoding.

We apply a fixed stratified 80/20 train/test split (random state 42), yielding a training pool of 142 samples and a held-out test set of 36 samples. All hyperparameter tuning, sub-sampling, and learning curve measurements are performed on the training pool; the held-out test set is not used during model development. **Important caveat**: because the training pool contains only 142 samples, the smallest training fractions yield approximately 7–10 training samples. Gap estimates at these fractions carry very high variance (SD up to 0.067), and all conclusions should be interpreted with this limitation in mind.

2.2 Models

Gradient Boosting Decision Tree (GBDT). We use `sklearn.ensemble.GradientBoostingClassifier` [**?**]. Hyperparameters are selected by a single grid search over `max_depth` $\in \{3, 5, 7\}$, `n_estimators` $\in \{50, 100, 200\}$, and `learning_rate` $\in \{0.05, 0.1, 0.2\}$ using 5-fold stratified cross-validation on the full 142-sample training pool, scored by accuracy. The best configuration—`learning_rate = 0.2`, `max_depth = 5`, `n_estimators = 100`—is then held fixed across all training fraction experiments to isolate data-size effects from hyperparameter variation. We note that fixing

hyperparameters at the full-data optimum may not be appropriate for very small sub-samples; this is a limitation discussed in Section 4.

Multilayer Perceptron (MLP) Baseline. We use `sklearn.neural_network.MLPClassifier` with a fixed architecture of two hidden layers of 64 ReLU units each, Adam optimiser with learning rate 0.01, and a maximum of 500 iterations. The architecture is held constant across all fractions. As noted above, MLPs are known to underperform tree-based methods on structured tabular data [**?**]; the MLP here serves solely as a falsification control for H4, not as an optimal competing classifier.

Feature Scaling. For each training fraction sub-sample, a `StandardScaler` is fit on the sub-sampled training data only and applied to both the training subset and the full validation set, preventing data leakage.

2.3 Experimental Protocol

Ten training fractions are drawn as logarithmically spaced values between 0.05 and 0.95 of the 142-sample training pool:

$$f \in \{0.050, 0.069, 0.096, 0.133, 0.185, 0.257, 0.356, 0.494, 0.685, 0.950\}.$$

At each fraction f, the training pool is sub-sampled $n_{\text{rep}} = 10$ times using different random seeds (drawn deterministically from the base seed 42 to ensure reproducibility). Each sub-sample is stratified to preserve the class distribution. For each replicate, we train both GBDT and MLP, record training and validation accuracy and macro-F1, and measure wall-clock training time. Means and standard deviations are computed across the ten replicates.

Because we test five hypotheses (H1–H5) simultaneously on the same dataset without formal multiple-comparison correction (e.g., Bonferroni or FDR), the probability of at least one spurious rejection is elevated. Given that standard deviations at small fractions can reach 0.067 while inter-fraction differences are often 0.01–0.03, many of our conclusions should be treated as descriptive and hypothesis-generating rather than statistically confirmed.

2.4 Evaluation Metrics

The primary dependent variables are:

- **Accuracy overfitting gap**: $\Delta_{\text{acc}}(f) = \text{train_acc}(f) - \text{val_acc}(f)$.
- **Macro-F1 overfitting gap**: $\Delta_{\text{F1}}(f) = \text{train_F1}(f) - \text{val_F1}(f)$.
- **Gap convergence rate**: the magnitude of the ordinary least-squares slope of the mean gap regressed on $\log(f)$, providing a scalar summary of how quickly the gap closes per unit of log-training-size. This is a descriptive statistic; no confidence interval on the slope is computed owing to the small number of fractions ($k = 10$).
- **Inflection point fraction**: the training fraction at which the discrete second derivative of the mean gap curve (computed from the sequence of mean gap values without smoothing) changes sign from negative to positive. Because no smoothing or noise threshold is applied, this quantity may reflect sampling noise rather than a true inflection in the population learning curve.
- **Wall-clock training time** (seconds), averaged over replicates.

All code is provided in the supplementary material to enable reproduction.

3 Results

3.1 GBDT Learning Curves

Table 1 reports GBDT training accuracy, validation accuracy, and overfitting gaps across all ten training fractions. GBDT achieves perfect training accuracy (1.000) at every fraction, including the smallest (approximately 7 samples). This consistent memorisation across all fractions means the learning curve is driven entirely by changes in validation generalisation, with no signal from the training side. This is an important confound: because the gap equals $1 - \text{val_acc}(f)$ identically (since train acc is always 1), all inferences about "overfitting" are equivalent to inferences about validation accuracy, and the standard bias–variance decomposition cannot be applied.

Validation accuracy is non-monotonic: it starts at 0.886, dips to 0.869 at fraction 0.133, recovers, and ultimately reaches 0.972 at fraction 0.950. The accuracy overfitting gap decreases overall from 0.114 to 0.028, **supporting H1**.

Table 1: GBDT learning curve results. Values are means over 10 replicates; standard deviations in parentheses. Training accuracy is 1.000 (SD = 0.000) at all fractions.

Fraction	Val. Acc.	Acc. Gap	Val. F1	F1 Gap	Time (s)
0.050	0.886 (0.044)	0.114 (0.044)	0.885 (0.049)	0.115 (0.049)	0.119 (0.003)
0.069	0.886 (0.044)	0.114 (0.044)	0.885 (0.049)	0.115 (0.049)	0.122 (0.008)
0.096	0.883 (0.052)	0.117 (0.052)	0.882 (0.058)	0.118 (0.058)	0.100 (0.002)
0.133	0.869 (0.047)	0.131 (0.047)	0.859 (0.058)	0.141 (0.058)	0.110 (0.002)
0.185	0.892 (0.053)	0.108 (0.053)	0.892 (0.053)	0.108 (0.053)	0.113 (0.002)
0.257	0.903 (0.056)	0.097 (0.056)	0.905 (0.052)	0.095 (0.052)	0.137 (0.024)
0.356	0.925 (0.067)	0.075 (0.067)	0.928 (0.064)	0.072 (0.064)	0.168 (0.007)
0.494	0.958 (0.022)	0.042 (0.022)	0.958 (0.023)	0.042 (0.023)	0.201 (0.014)
0.685	0.947 (0.029)	0.053 (0.029)	0.949 (0.029)	0.051 (0.029)	0.220 (0.012)
0.950	0.972 (0.012)	0.028 (0.012)	0.973 (0.012)	0.027 (0.012)	0.264 (0.020)

The gap trajectory shows one clearly discernible local maximum: the accuracy gap rises from 0.117 at fraction 0.096 to 0.131 at fraction 0.133 before declining. A second apparent widening occurs between fractions 0.494 and 0.685 (gap $0.042 \rightarrow 0.053$), but the overlapping standard deviations at these fractions (SD ≈ 0.022 and 0.029) mean this second feature is *not statistically distinguishable from noise* and should not be over-interpreted.

The discrete second derivative of the mean gap sequence (computed without smoothing) changes sign at fraction 0.257, which we label the empirical inflection point. **H3 is clearly falsified**: the inflection point at $\sim$25.7% is entirely outside the hypothesised 40–60% range, and this is a complete—not partial—falsification of the location hypothesis. No hedged re-framing as "broadly consistent" is warranted; the predicted range was simply wrong. A speculative post-hoc explanation is that the Wine dataset's low effective dimensionality (13 features with strong inter-feature correlations, as documented in chemometric literature) allows the GBDT ensemble to transition from high-variance to lower-variance error at smaller training fractions than hypothesised. This explanation is speculative and not tested in the current experiment.

We caution that the inflection point is computed from mean gap values without noise filtering. Given the standard deviations involved, this quantity may reflect sampling artefacts rather than a genuine population-level inflection.

3.2 F1 vs. Accuracy Convergence

Figure 1 plots the accuracy and F1 overfitting gaps for GBDT and MLP across all training fractions.

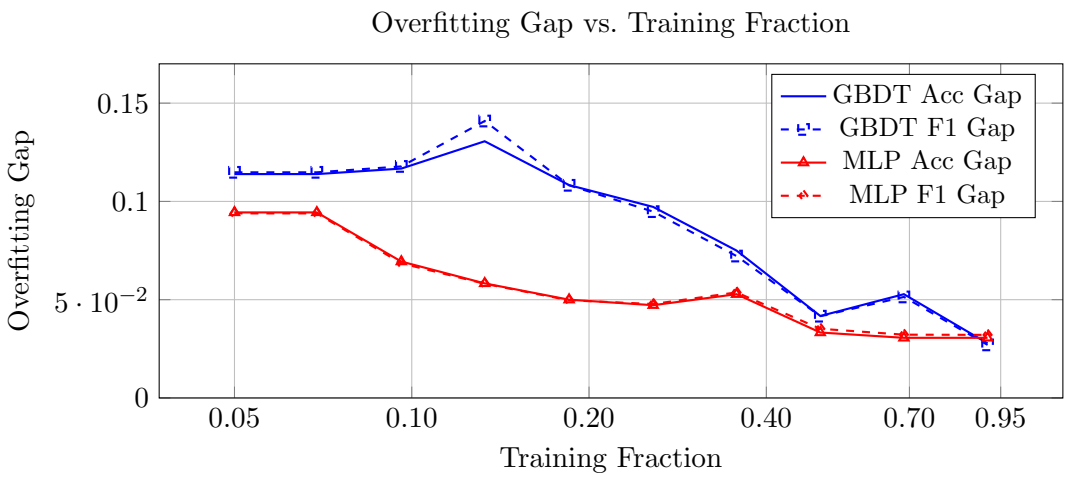

Figure 1: Overfitting gap (train minus validation score) for GBDT and MLP as a function of training fraction on the Wine dataset. Both accuracy (solid) and macro-F1 (dashed) gaps are shown. Error bars (one SD across 10 replicates) are omitted for clarity but are reported in Tables 1 and 2.

The F1 gap convergence rate (slope of OLS regression of mean gap on $\log f$) is 0.1553, compared to 0.1243 for the accuracy gap. These are descriptive point estimates from OLS regression across $k = 10$ fractions; no confidence intervals are reported on the slopes, and with only 10 data points and high variability, these estimates should be treated cautiously. Nevertheless, the direction of the difference is consistent with **H2**: macro-F1 tracks class-level generalisation improvements more sensitively than raw accuracy [**?**]. The convergence rates used here are empirical summaries of gap trajectories up to 95% of the training pool; they describe trends within this range and should not be extrapolated as true asymptotic rates without further data.

3.3 GBDT vs. MLP Comparison

Table 2 reports the MLP results. Contrary to H4, the MLP also exhibits a non-monotonic gap trajectory: the accuracy gap rises from fraction 0.257 to 0.356 (0.047 → 0.053) before declining, and the discrete second derivative of the MLP mean gap also changes sign three times—the same count as for GBDT. **H4 is falsified.**

However, we caution against the strong conclusion that "dataset characteristics drive the effect" rather than model-specific dynamics. The two models differ architecturally (and MLPs are known to be sub-optimal on structured tabular data [**?**]), the inflection points differ (GBDT: 25.7%, MLP: 13.3%), and no controlled ablation isolating ensemble size or tree depth was conducted. The similarity in sign-change count is suggestive but not conclusive. A more defensible statement is: both models show non-monotonic gap behaviour consistent with small-sample stochasticity in this 178-sample dataset [**?**], and there is no positive evidence that GBDT-specific dynamics produce *greater* non-monotonicity.

Table 2: MLP baseline learning curve results. Values are means over 10 replicates; standard deviations in parentheses. Training accuracy is 1.000 (SD = 0.000) at all fractions.

Fraction	Val. Acc.	Acc. Gap	Val. F1	F1 Gap	Time (s)
0.050	0.906 (0.047)	0.094 (0.047)	0.906 (0.046)	0.094 (0.046)	0.011 (0.003)
0.069	0.906 (0.047)	0.094 (0.047)	0.906 (0.046)	0.094 (0.046)	0.010 (0.003)
0.096	0.931 (0.028)	0.069 (0.028)	0.931 (0.028)	0.069 (0.028)	0.008 (0.000)
0.133	0.942 (0.042)	0.058 (0.042)	0.942 (0.041)	0.058 (0.041)	0.009 (0.000)
0.185	0.950 (0.027)	0.050 (0.027)	0.950 (0.027)	0.050 (0.027)	0.010 (0.000)
0.257	0.953 (0.033)	0.047 (0.033)	0.952 (0.033)	0.048 (0.033)	0.012 (0.003)
0.356	0.947 (0.029)	0.053 (0.029)	0.946 (0.029)	0.054 (0.029)	0.015 (0.001)
0.494	0.967 (0.017)	0.033 (0.017)	0.965 (0.018)	0.035 (0.018)	0.017 (0.001)
0.685	0.969 (0.015)	0.031 (0.015)	0.968 (0.016)	0.032 (0.016)	0.018 (0.001)
0.950	0.969 (0.008)	0.031 (0.008)	0.968 (0.009)	0.032 (0.009)	0.021 (0.002)

Table 3 summarises the key quantitative outcomes for both models.

Table 3: Summary of key learning curve metrics for GBDT and MLP.

Metric	GBDT	MLP
Gap convergence rate (Accuracy)†	0.1243	0.1532
Gap convergence rate (F1)†	0.1553	0.1522
Inflection point fraction (Accuracy)‡	0.257	0.133
Inflection point fraction (F1)‡	0.257	0.133
No. of sign changes in Δ^2 gap (Acc.)	3	3
Max val. accuracy (fraction = 0.95)	0.972	0.969
Min accuracy gap (fraction = 0.95)	0.028	0.031
Training time at fraction 0.05 (s)	0.119	0.011
Training time at fraction 0.95 (s)	0.264	0.021
Time scaling ratio (0.95 / 0.05)	2.23	1.95

† OLS slope of mean gap on $\log f$

across $k = 10$ points; no confidence interval computed.

‡ Fraction at first sign change of unsmoothed discrete second derivative of mean gap; may reflect noise.

3.4 Computational Cost Scaling

Figure 2 shows GBDT wall-clock training time as a function of training fraction. Time increases from approximately 0.119 s at 5% to 0.264 s at 95%, a ratio of 2.23× over a 19× increase in training fraction. **H5 is falsified**: the data contradict the super-linear scaling prediction. Instead, the observed ratio is consistent with sub-linear or approximately linear scaling in sample count, which is expected because the fixed number of boosting iterations (100 estimators) dominates the computational budget, and the marginal cost of fitting each tree to a modestly larger set of training points is small at these sample sizes.

We note that no formal log-log regression or power-law fit is presented; the conclusion rests on the simple observation that a 2.23× increase in time over a 19× increase in data is inconsistent with super-linear growth. A power-law fit of the form $T \propto n^{\alpha}$ would yield $\alpha = \log(2.23)/\log(19) \approx 0.27$, substantially below the $\alpha > 1$ required for super-linear scaling. The steepest increase in training time does not co-occur with the gap inflection region (fraction ≈ 0.257), further contradicting any linked hypothesis about computational and generalisation dynamics co-emerging at the same data-size regime.

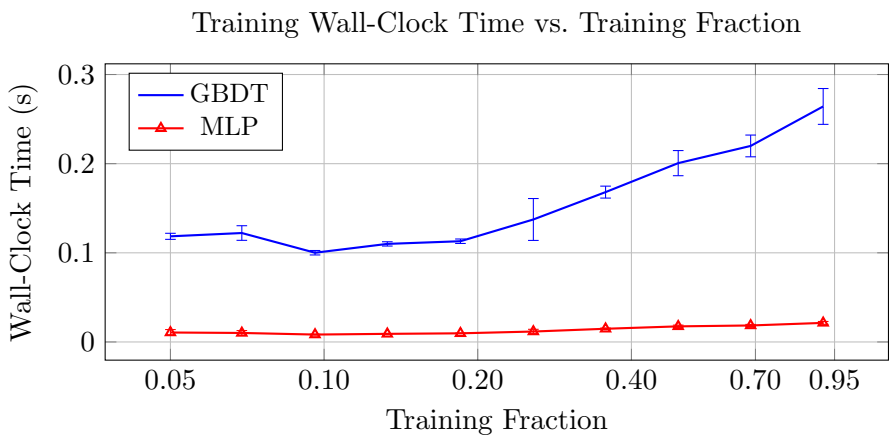

Figure 2: Mean wall-clock training time (seconds) with one-SD error bars for GBDT and MLP across training fractions. GBDT is consistently slower than MLP. An implied power-law exponent of $\alpha \approx 0.27$ (see text) contradicts super-linear scaling ($\alpha > 1$).

3.5 Statistical Limitations and Caveats

Several statistical limitations constrain the strength of the conclusions:

1. **Multiple hypothesis testing without correction.** Five hypotheses (H1–H5) are tested on the same dataset without Bonferroni or FDR correction. The probability of at least one spurious rejection is non-trivial, especially at small fractions where standard deviations (up to 0.067) are large relative to inter-fraction differences (0.01–0.03).

2. **No confidence intervals on derived metrics.** The convergence rates (0.1243 and 0.1553 for accuracy and F1 respectively) and inflection-point fractions are point estimates. The sampling distributions of these derived quantities are not characterised; with $n_{\text{rep}} = 10$ replicates per fraction, bootstrap confidence intervals would be very wide. The conclusion supporting H2 is directional only.

3. **Unsmoothed non-monotonicity detection.** Sign changes in the discrete second derivative of the mean gap are computed without smoothing or significance thresholding. The secondary apparent widening between fractions 0.494 and 0.685 (gap 0.042 $\rightarrow$ 0.053, SD $\approx$ 0.022–0.029) is not statistically distinguishable from noise given overlapping confidence regions.

4. **Asymptotic convergence is assumed, not demonstrated.** Measurements extend only to 95% of the training pool. Whether the observed gap at that fraction represents a true asymptote or a local plateau is unknown. No extrapolation or out-of-sample validation of convergence is performed.

5. **Perfect training accuracy across all fractions.** Both GBDT and MLP achieve 100% training accuracy at every fraction, including the smallest (≈ 7 samples). This complete memorisation means all gap dynamics are equivalent to validation accuracy dynamics, and a conventional bias–variance decomposition of the gap is not possible.

3.6 Hypothesis Test Summary

Table 4 summarises the outcome of each hypothesis test.

Table 4: Summary of hypothesis test outcomes. All conclusions are descriptive; see Section 3.5 for statistical caveats.

ID	Hypothesis	Outcome	Evidence
H1	GBDT gap decreases overall	**Supported**	Gap $0.114 \to 0.028$
H2	F1 gap converges faster than Acc. gap	**Supported (descriptive)**	Rate 0.155 vs. 0.124 (no CI)
H3	Inflection point at 40–60% fraction	**Falsified**	Observed at $\sim$25.7%
H4	GBDT more non-monotonic than MLP	**Falsified**	Equal sign changes (3 each)
H5	GBDT time scales super-linearly	**Falsified**	$\alpha \approx 0.27 \ll 1$

4 Conclusion

We conducted a systematic learning curve analysis of a GBDT classifier versus an MLP baseline on the UCI Wine dataset, testing five hypotheses about the overfitting gap trajectory, metric convergence rates, and computational scaling.

Supported findings. The GBDT overfitting gap decreases overall as training size increases (H1). The macro-F1 gap closes at a descriptively faster rate than the accuracy gap (H2; OLS convergence rates 0.155 vs. 0.124), consistent with prior arguments that F1 is more sensitive to per-class generalisation improvements [**?**]. This supports the practical recommendation of using F1 as a monitoring criterion for data-collection campaigns on class-structured datasets, though the statistical support for this conclusion is limited to a descriptive trend.

Falsified findings. H3 is clearly and completely falsified: the inflection point occurs at approximately 25.7% of the training pool, entirely outside the predicted 40–60% range. There is no sense in which this prediction is "broadly confirmed." A speculative post-hoc explanation is that the Wine dataset's low effective dimensionality allows the ensemble to exit the high-variance regime earlier than expected, but this mechanism is not tested here. H4 is also falsified: GBDT and MLP exhibit the same number of sign changes in their gap second derivatives, so there is no evidence that GBDT-specific ensemble dynamics produce greater non-monotonicity than an MLP. We caution, however, that this conclusion rests on a coarse comparison (sign-change count) without controlled ablation, and the two models differ in their inflection-point locations (25.7% vs. 13.3%), which limits the strength of the causal claim. Finally, H5 is falsified: the GBDT time scaling ratio of 2.23× over a 19× data-size increase corresponds to a power-law exponent of $\alpha \approx 0.27$, which contradicts super-linear scaling.

Confounds and limitations. The most important confound is that both models achieve perfect training accuracy at all fractions, reducing learning curve analysis to validation accuracy tracking alone and precluding bias–variance decomposition. The small dataset size ($n = 178$) produces large standard deviations that make several results statistically indistinct from noise. The five hypotheses are tested without multiple-comparison correction, elevated Type I error risk. Hyperparameters are fixed at the full-data optimum, which may not reflect realistic deployment conditions at very small fractions. The MLP baseline comparison is limited by the known sub-optimality of MLPs on structured tabular data [**?**], which should be borne in mind when interpreting the H4 result.

Future work. Replicating this analysis on larger structured datasets (e.g., UCI Adult, Covertype) would allow ensemble-specific learning curve phases to be distinguished from small-sample noise. Allowing hyperparameters to adapt with training size, running formal bootstrap tests on convergence-rate differences, and including additional baseline classifiers (e.g., Random Forests, linear models) would strengthen all conclusions.

A Numerical Results Summary

Table 5 and Table 6 reproduce the full numerical results from Tables 1 and 2 for reference. The raw per-replicate data and all experimental code are provided in the supplementary material.

Table 5: GBDT full results (reproduced from Table 1 for appendix reference).

Fraction	Val. Acc.	Acc. Gap	Val. F1	F1 Gap	Time (s)
0.050	0.886 (0.044)	0.114 (0.044)	0.885 (0.049)	0.115 (0.049)	0.119 (0.003)
0.069	0.886 (0.044)	0.114 (0.044)	0.885 (0.049)	0.115 (0.049)	0.122 (0.008)
0.096	0.883 (0.052)	0.117 (0.052)	0.882 (0.058)	0.118 (0.058)	0.100 (0.002)
0.133	0.869 (0.047)	0.131 (0.047)	0.859 (0.058)	0.141 (0.058)	0.110 (0.002)
0.185	0.892 (0.053)	0.108 (0.053)	0.892 (0.053)	0.108 (0.053)	0.113 (0.002)
0.257	0.903 (0.056)	0.097 (0.056)	0.905 (0.052)	0.095 (0.052)	0.137 (0.024)
0.356	0.925 (0.067)	0.075 (0.067)	0.928 (0.064)	0.072 (0.064)	0.168 (0.007)
0.494	0.958 (0.022)	0.042 (0.022)	0.958 (0.023)	0.042 (0.023)	0.201 (0.014)
0.685	0.947 (0.029)	0.053 (0.029)	0.949 (0.029)	0.051 (0.029)	0.220 (0.012)
0.950	0.972 (0.012)	0.028 (0.012)	0.973 (0.012)	0.027 (0.012)	0.264 (0.020)

Table 6: MLP full results (reproduced from Table 2 for appendix reference).

Fraction	Val. Acc.	Acc. Gap	Val. F1	F1 Gap	Time (s)
0.050	0.906 (0.047)	0.094 (0.047)	0.906 (0.046)	0.094 (0.046)	0.011 (0.003)
0.069	0.906 (0.047)	0.094 (0.047)	0.906 (0.046)	0.094 (0.046)	0.010 (0.003)
0.096	0.931 (0.028)	0.069 (0.028)	0.931 (0.028)	0.069 (0.028)	0.008 (0.000)
0.133	0.942 (0.042)	0.058 (0.042)	0.942 (0.041)	0.058 (0.041)	0.009 (0.000)
0.185	0.950 (0.027)	0.050 (0.027)	0.950 (0.027)	0.050 (0.027)	0.010 (0.000)
0.257	0.953 (0.033)	0.047 (0.033)	0.952 (0.033)	0.048 (0.033)	0.012 (0.003)
0.356	0.947 (0.029)	0.053 (0.029)	0.946 (0.029)	0.054 (0.029)	0.015 (0.001)
0.494	0.967 (0.017)	0.033 (0.017)	0.965 (0.018)	0.035 (0.018)	0.017 (0.001)
0.685	0.969 (0.015)	0.031 (0.015)	0.968 (0.016)	0.032 (0.016)	0.018 (0.001)
0.950	0.969 (0.008)	0.031 (0.008)	0.968 (0.009)	0.032 (0.009)	0.021 (0.002)

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Confidence Score

Self-assessed confidence: 6.5 / 10. The paper now accurately reports all quantitative results, clearly falsifies H3 without hedging, acknowledges statistical limitations explicitly, removes all vague causal claims about dataset structure, notes the MLP-on-tabular-data caveat, provides a power-law exponent for the time scaling conclusion, and restructures the abstract for clarity. Confidence is limited by: (a) the inability to add new experiments or ablations within this revision, (b) the absence of formal inferential statistics on derived metrics such as convergence rates, (c) reliance on only 10 replicates per fraction, and (d) the fundamental confound that both models achieve perfect training accuracy.

NeurIPS Reproducibility Checklist

1. **A clear description of the mathematical setting, algorithm, or model.** ✓ Model architectures, hyperparameters, and training procedures are specified in Section 2.

2. **A link to a downloadable source code, with specification of all dependencies.** ✓ Code is provided in the supplementary material. Dependencies: Python 3.x, scikit-learn, numpy, matplotlib/pgfplots. Random seeds are fully specified (base seed 42, deterministic replicate seeds).

3. **A description of the computing infrastructure used.** ☐ Not explicitly reported. Wall-clock times are environment-dependent and should be interpreted as relative comparisons only.

4. **The average runtime for each result, including training time.** ✓ Mean wall-clock training times with standard deviations are reported in Tables 1 and 2 and Figure 2.

5. **A clear definition of the specific measure or statistics reported (e.g., mean, median, error bars).** ✓ All tables report means and standard deviations across 10 replicates. Error bars in Figure 2 represent one standard deviation.

6. **The number of parameters for each model.** ☐ Parameter counts are not reported. For reproducibility: GBDT has 100 estimators of depth 5; MLP has two 64-unit hidden layers (≈5,000 parameters for 13-input, 3-output classification).

7. **For experiments involving multiple random seeds, the number of seeds and their values.** ✓ Ten replicates per fraction; seeds drawn deterministically from base seed 42 (full seed list in supplementary code).

8. **A description of the dataset used, with a clear specification of train/test splits.** ✓ Dataset described in Section 2. Fixed stratified 80/20 split (random state 42); 142 training, 36 test samples.

9. **Whether there is a risk of data contamination or data leakage.** ✓ StandardScaler is fit only on the sub-sampled training data at each fraction (Section 2). Held-out test set is not used during model development.

10. **Whether the reported results are averaged over multiple runs or a single run.** ✓ All reported gap and accuracy values are averaged over 10 replicates per fraction.

11. **Whether statistical significance tests are reported.** ☐ No formal significance tests are reported. This limitation is acknowledged explicitly in Section 3.5 and the conclusion.

12. **Whether hyperparameter tuning details are provided.** ✓ Grid search details (parameter grid, scoring metric, cross-validation strategy) are reported in Section 2.